

Heat loss and insulation

Sometimes we wish to produce as much heat as we can, rather than as little as possible. When we do, it is important to prevent the heat escaping. The main ways to reduce heat loss from buildings are to keep doors and windows closed and to install heat insulation.

▷ Keeping warm for less

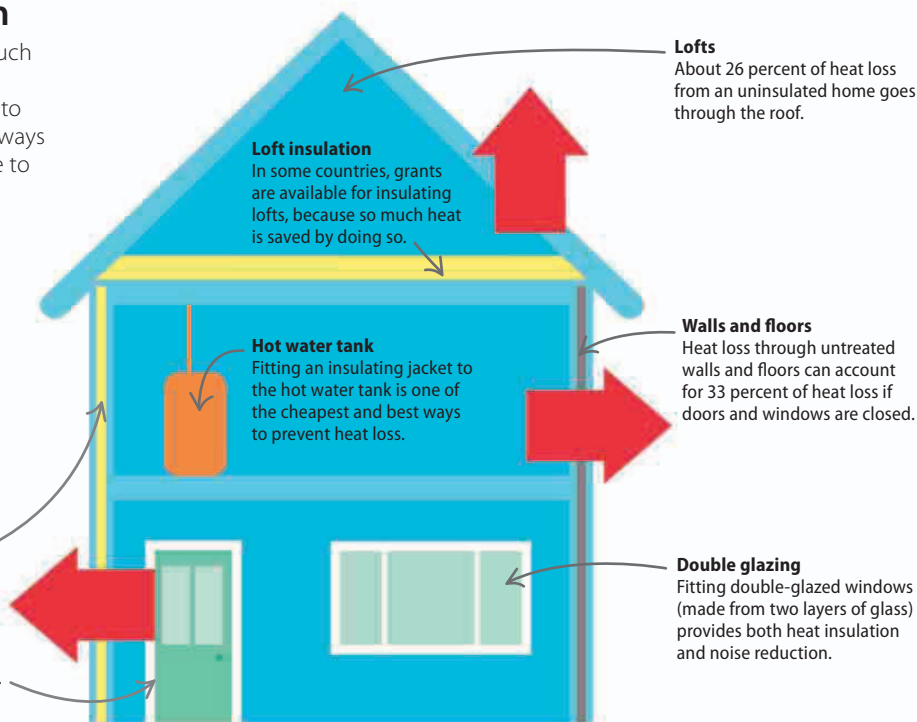
Heating a well-insulated home costs only a small fraction of the amount required to heat an uninsulated one. Here are some ideas to help keep a house warm and save money.

Cavity wall insulation

Most houses are built with hollow outer walls. The gaps can be filled with foam, which sets hard and provides effective insulation.

Doors and windows

In an uninsulated house, gaps around doors and windows can account for 11 percent of heat loss.

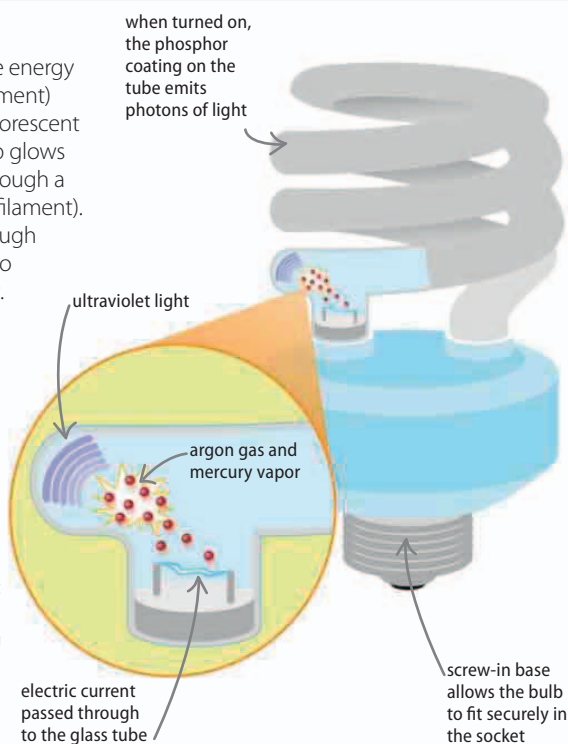


Fluorescent bulbs

One of the easiest ways to save energy is to replace incandescent (filament) bulbs, with energy-efficient fluorescent versions. An incandescent bulb glows because the current passes through a high-resistance bare wire (the filament). The resistance means that enough electrical energy is converted to heat to make it glow with light.

▷ Compact fluorescent lamps (CFL)

Compact fluorescent lamps (CFLs) are gradually replacing domestic, incandescent bulbs because they are more energy efficient and last longer. CFLs feature a spiraling glass tube full of gases, which emit ultraviolet light when an electric current passes through them. This triggers a phosphor coating on the tube to shine brightly.



▽ Compact fluorescent lamp

If 100 J (joules) of energy is passed through a CFL, most of the energy appears as light (75 J), so less electricity is required and less unnecessary heat (25 J) is produced.



▽ Domestic incandescent bulb

From 100 J of electrical energy, most of the energy supplied to a domestic incandescent bulb is converted to heat (90 J), while only a small portion (10 J) is converted to light.

